

HACK-O-WEEN

TEAM NAME: The Cursed Coders

TRACK SELECTED 2: "Phantom Predictions
(AI for Good & Predictions)"

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PROBLEM STATEMENT

Develop an AI system to predict disease outbreaks early, enabling timely intervention to reduce health and economic impacts.

Why relevant?

- Early prediction saves lives and resources.
- Traditional reporting is too slow for rapid action.

Objectives:

- Predict outbreak risk before escalation.
- Enable timely response by health organizations.

SOLUTION

Solution Overview :

- AI processes health records and symptoms to predict probable outbreak zones.
- Presents risk as interactive maps and real-time alerts.
- Combines supervised learning (Random Forest, LSTM) with time-series analysis.

SOLUTION

Why This Solution Works?

- Integrates multi-source data (health, location, symptoms, environment).
- ML adapts as new data emerges.
- Interactive dashboard for quick action, easy interpretation.

SOLUTION

How Solution Works:

- Step 1: Collect health and environment data (CSV, APIs, public datasets).
- Step 2: Preprocess and feature engineer inputs.
- Step 3: Train ML model to classify/predict outbreaks.
- Step 4: Serve results via backend API.
- Step 5: Display on dashboard map + alert policy makers.

TECK STACK

- FRONTEND: React, Chart.js, Mapbox, JavaScript, HTML, CSS
- BACKEND: Python, Flask, FastAPI, Pandas, Scikit-learn
- DATABASE: SQLite, PostgreSQL
- APIs/SERVICES: RESTful API endpoints, Mapbox API, Public datasets APIs

BUSINESS SCOPE

Value to companies, NGOs, and organizations:

- Enables faster, data-driven decisions for health resource allocation, outbreak containment, and public safety, benefiting hospitals, non-profits, insurance companies, and government agencies.

Feasibility:

- High, due to readily available public health datasets, mature open-source ML tools, and free or low-cost cloud hosting options, with scalability for real-world deployment.

Business applications:

- Used for epidemic forecasting, healthcare resource planning, insurance risk management, pharmaceutical demand prediction, and disaster response optimization.